# Practice Quiz 10D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

*Repairing an incomplete induction argument.*

1. Let $a_1 = 1$, $a_2 = 2$, and

$$a_n = a_{n-1} + 2a_{n-2} \qquad (n \geq 3).$$

A student claims $a_n = 2^{n-1}$ for every integer $n \geq 1$ and writes:

> The formula holds at $n = 1$. Assume $a_n = 2^{n-1}$. Then
>
> $$a_{n+1} = a_n + 2a_{n-1} = 2^{n-1} + 2 \cdot 2^{n-2} = 2^n.$$
>
> Thus induction proves the formula.

(a) Identify the missing base case and the unsupported use of the inductive hypothesis.

**Solution** The proof must also check $n = 2$, because the recurrence is valid only from index 3 onward. At $n = 1$ its displayed step would use an undefined term $a_0$. Also, assuming only $a_n = 2^{n-1}$ does not justify the substitution $a_{n-1} = 2^{n-2}$. The hypothesis must cover both earlier terms used in the recurrence. ∎

(b) Give a corrected proof using strong induction.

**Solution** Let $m \geq 1$ and assume

$$a_k = 2^{k-1} \quad \text{for every integer } k \text{ with } 1 \leq k < m.$$

For $m = 1$, $a_1 = 1 = 2^0$. For $m = 2$, $a_2 = 2 = 2^1$.

For $m \geq 3$, both $m - 1$ and $m - 2$ lie between 1 and $m - 1$, so the hypothesis applies at both indices. Hence

$$a_m = a_{m-1} + 2a_{m-2} = 2^{m-2} + 2 \cdot 2^{m-3} = 2^{m-1}.$$

The base cases and induction step are complete. By strong induction, $a_m = 2^{m-1}$ for every integer $m \geq 1$. ∎

*Solutions revised using GPT-6 Astra in the Codex harness.*